

StressBeat: Academic Stress vs. Physiological Recovery

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RESEARCH QUESTION

How does the return to academic rigor following Spring Break impact physiological recovery, as measured by morning Heart Rate Variability (RMSSD)?

INTRODUCTION

Motivation

- Students often experience significant physiological and psychological stress fluctuations during the transition from holiday periods back to university life
- RMSSD (Root Mean Square of Successive Differences) is a primary metric of parasympathetic activity. Lower RMSSD values typically indicate high stress and inadequate recovery.

This study utilizes established HRV analysis models to quantify the impact of academic workload on the autonomic nervous system.

MATERIALS AND METHODS

Instrumentation & Objective Metric

- Data was acquired using the Polar H10 chest strap. As a bipolar ECG sensor, it records electrical activity via conductive textile electrodes at a 1000Hz sampling rate, providing clinical-grade fidelity for R-R interval detection compared to optical sensors.
- The primary objective measure RMSSD is calculated as:

$$RMSSD = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N-1} (RR_{i+1} - RR_i)^2}$$

- A daily Academic Stress Survey was also implemented. Using a 1-10 scale (1: Relaxed to 10: Extreme Pressure), perceived mental workload was quantified to correlate with heart rate variability trends.

Protocol

- Tracking spanned 6 weeks (Mar 8 – Apr 19). Every morning upon waking, a 2:30 minute resting ECG was recorded using the Polar H10 while simultaneously completing the stress survey.
- Raw R-R intervals were processed in R to:
 - Extract daily RMSSD and subjective stress scores.
 - Perform a Pearson Correlation test to evaluate the relationship between stress and recovery.

RESULTS

Results

- 41 valid data points were collected over 43 days, covering both vacation and high-stress academic weeks.
- RMSSD dropped from ~70ms (Vacation) to ~45ms (High-Load), showing clear physiological suppression during deadlines.
- Recovery began declining 24–48 hours before major deadlines, reflecting anticipatory stress
- Two artifact (3/26 & 4/5) was flagged and excluded to ensure baseline integrity
- Statistical Analysis (Spearman's Correlation)
 - $rs = -0.604$, $p < 0.001$
 - A strong negative correlation exists; higher academic stress significantly reduces parasympathetic recovery.
 - The $p < 0.001$ result validates RMSSD as a reliable biomarker for monitoring student burnout risk.

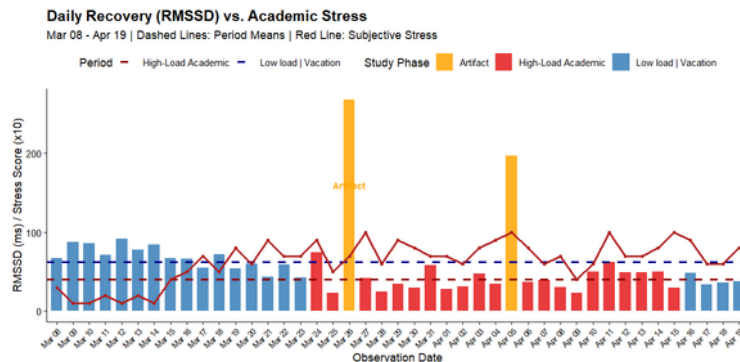


Figure 1. Daily Recovery (RMSSD) vs. Academic Stress

Recovery Heatmap (Morning RMSSD)

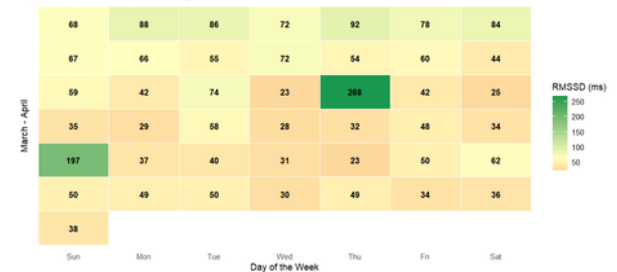


Figure 2. Recovery Heatmap

Statistical Correlation Analysis

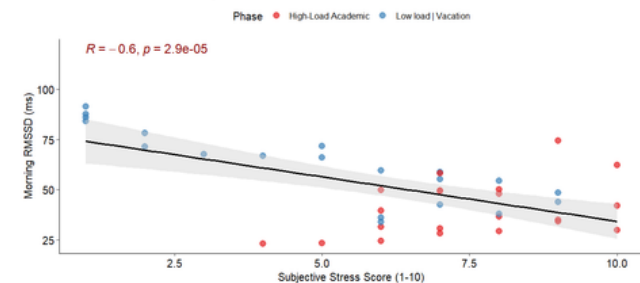


Figure 3. Statistical Correlation

DISCUSSION

Objective & Takeaways

- The project confirmed that academic pressure physically suppresses autonomic recovery. Data shows subjective stress and RMSSD are inversely linked, making burnout physiologically measurable.

Limitations

- This N=1 study is personalized and not generalizable. Uncontrolled factors like caffeine and exercise, along with occasional sensor artifacts, influenced the data.

Future Improvements

- Future versions should include sleep and diet tracking for higher accuracy. Testing a larger cohort of students would allow for a broader comparison of stress resilience.

REFERENCES

